Hand-in assignment 2 solutions

Please note that these solutions are only meant to serve as a guide, the questions may have a number of alternative acceptable solutions. The advise is: **If you do not get full mark for a question, try to do it again independenly after going through this document**.

Solutions

1. (a) True. Let (x_n) be a sequence in X such that (x_n) converges to x, since X is finite we have that sequence (x_n) is eventually constant and so $(f(x_n))$ is also eventually constant. We then have that $(f(x_n))$ converges to $f(x)$ and so f is continuous.

   (b) False. Let $(X, d_X) = (\mathbb{R}, d_D)$ and $(Y, d_Y) = (\mathbb{R}, d_E)$, the function $f : X \to Y$ defined by $f(x) = x$ is a continuous bijection but $f^{-1} : Y \to X$ is not continuous (why?).

   (c) False. Consider $(\mathbb{R}, d_E)$ and $(\mathbb{R}^2, d_E)$, the map $f : \mathbb{R} \to \mathbb{R}^2$ defined by $f(x) = (x, 0)$ is an isometry but not surjective.

   (d) False. Let X be a set with at least two elements and Y be a one element set, and consider discrete metrics on both sets. The unique function $f : X \to Y$ is Lipschitz continuous, but not injective.

   (e) True. Suppose $f : X \to Y$ is Lipschitz continuous and let $\varepsilon > 0$, take $\delta = \frac{\varepsilon}{k}$, we then have that for each $x_1, x_2 \in X$ and $d_X(x_1, x_2) < \delta$, we have $d_Y(f(x_1), f(x_2)) \leq k d_X(x_1, x_2) < k \cdot \frac{\varepsilon}{k} = \varepsilon$.

2. Let $X = \{0, 1\}$. The list of topologies on X are: (Make sure you are convinced that these are the only possibilities, note that $PP(X)$ has 16 elements).

   (a) $\tau_1 = \{\emptyset, X\}$

   (b) $\tau_2 = \{\emptyset, \{0\}, X\}$

   (c) $\tau_3 = \{\emptyset, \{1\}, X\}$

   (d) $\tau_4 = P(X)$

   (X, τ_2) and (X, τ_3) are homeomorphic with homeomorphism $f : X \to X$
   defined by $f(0) = 1$ and $f(1) = 0$. It is easy to check that f is a continuous bijection with f^{-1} also continuous. (Easy, again there are no other possibilities).